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Applicant: Huawei Technologies Co., Ltd.
Our Ref.: P61434/WOEP

In response to the communication pursuant to Rules 161(1) and 162 EPC of October 11, 2023.

The applicant wishes to **proceed further** with the present application.

As a response, we file amended claims 1 – 13 and amended description pages 3, 17, 18. We comment on the remarks of the examiner as follows:

1. Amendments and Disclosure

A new independent claim 1 was drafted based upon the features of previous claims 1, 9 and 10. The new feature combination is readily disclosed by the references of the dependent claims.

Moreover, independent claim 1 was amended by adding that the multiple aggregate functions correspond to multiple best-response policies for the agents. This amendment was requested by the examiner. It is readily disclosed by p. 3, l. 21 of the original application documents.

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Furthermore, independent claim 1 was amended by defining that the two-agent system is a two-player zero-sum game. This amendment was requested by the examiner. Since the feature was considered an essential feature by the examiner, we believe disclosure is not an issue.

The new independent claim 1 was kept in the one-part form, since a wording in the two-part form would lead to very lengthy and unintelligible claims. The examiner is most respectfully requested to accept the one-part form in this case.

The amended description pages 3, 17, 18 replace the previous description pages 3, 17 and 18. The state of the art documents were cited. The description was brought into line with the amended claims according to the case law T18/1989. The last paragraph of the description has been deleted so as to bring the description into conformity with European patent practice.

2. Clarity

In section 2.1, the examiner objects that the term "two-agent input system" being defined as "comprising a definition of a two-agent system and defining behaviour patterns of two agents based on system states" were vague and unclear. We put forward that especially with the definition of the two-agent system as a two-agent zero-sum game, the objected to terms are now perfectly clear. The feature clearly shows that the definition of the game is part of the two-agent system input. Also, the behavioural patterns of the agents are perfectly clear to the skilled person, worst in game theory.

The examiner argues that it were not clear what qualifies as a definition of a two-agent system. Since it is now defined that two-agent system is a zero-sum game, it is perfectly clear that the definition is the definition of the zero-sum game. Moreover, the examiner argues that it were not clear what qualified as the behaviour patterns of two agents based on the system states. Also this is perfectly clear from the wording of the feature itself. The behaviour of the individual agent based upon the encountered system state is meant. This is perfectly clear from the claim wording. We believe that at least with the amendments, the clarity objection is readily overcome.

Moreover, in section 2.2 the examiner objects that the term "an indication of an input system state" were unclear. We believe this clarity objection is now justified. An indication of an

input system state is readily understandable to the skilled person. It means an information or notification regarding the system state that is used as an input, as a starting position. We do not see how a person skilled in the art of game theory can misunderstand this.

Furthermore, in section 2.3, the examiner objects to the term "aggregate functions". Especially, the examiner asks what constituted an aggregate function. Also here, this is a term perfectly clear in the area of game theory. Asking the applicant to define trivial features put some undue burden on the applicant, and also leads to lengthy and unintelligible claims. In the respective section of the international search report, the examiner suggests that p. 4, l. 4 – 5 might overcome this objection. We put forward, that the respective feature has been added. We believe that at least with the amendment, the clarity objection is readily overcome.

Also, in section 2.4, the examiner objects that the relationship between the partially optimized outputs and the multiple aggregate functions were not clear. Also this is in our opinion perfectly clear. The multiple at least partially optimized outputs are the outputs of the multiple aggregate functions. This is perfectly clear from the claim wording. We do not see how the skilled person can misunderstand this.

Moreover, in section 2.6, the examiner objects that it were not clear how to assess whether the multiple aggregate functions are determined in a single iteration. In our opinion, this clarity objection is now justified. The inventive method does not deal with determining <u>if</u> the result is achieved in one iteration. This question is not relevant to the present invention. We do not see how this can warrant a clarity objection.

Furthermore, in section 2.7, the examiner objects that the scope of the protection is broader than justified, since it were not defined that the system is zero-sum game. We put forward that this objection is readily overcome by performing the above-specified amendment. The system has now been defined as a zero-sum game.

Finally, in sections 2.5 and 2.8 – 2.17, the examiner suggests that the features of all claims 1 – 9 would define the invention not by technical means, but by results to be achieved. We do not agree with this assessment. In all of the claims, specific technical means and processing steps are defined. Especially though, the examiner does not object to previous dependent claim 10 in this regard. Since the features of this claim are now present in independent claim 1, specific technical features, which seem to satisfy the examiner's requirement in this regard are now present in independent claim 1, and therefore also by means of reference, in all other

dependent claims. At least with this amendment, we believe that the objection is readily overcome.

3. Novelty

The examiner did not object to a lack of novelty of previous dependent claim 10. Since the features of this claim are present in the independent claim now, we believe that novelty is readily given and therefore refrain from further elaborations in this regard.

4. Inventive step

4.1 Closest state of the art

We agree with the examiner that document D1 should be considered as the closest state of the art document on file. Document D1 is a scientific paper dealing with game theory. Especially, determining nash equilibria is shown.

With regard to the disclosure of D1, we agree with the examiner. Moreover, we agree with the examiner that D1 does not show the features of previous dependent claim 10. Especially, D1 does not show that the set of random system states are generated based on a predetermined probability distribution.

This distinguishing feature has the objective technical effect of achieving a simple system set up. The teaching of D1 is therefore disadvantageous with regard to a complicated system set up.

4.2 Problem

The above-mentioned disadvantages of the closest state of the art require an improvement with regard to a simplified system set up.

4.3 Solution according to the present invention

The above-mentioned problem is solved by the present invention according to the independent claims. Especially, the problem is solved by the set of random system states

being generated based upon a predetermined probability distribution. This allows for a simple system set up while achieving reliable results.

4.4 Invention is not rendered obvious

4.4.1 D1 alone

The present invention according to the new independent claims is not rendered obvious by D1 alone since D1 does not go into detail how the system is set up. Especially, how the initial states are set, is not shown by D1. Most importantly though, document D1 does not show or hint at a set of random system states being generated based upon a predetermined probability distribution.

Inventive step over D1 alone is therefore readily given.

4.4.2 D1 + remaining state of the art

Also a combination of D1 and any or all of the remaining state of the art documents fails to render the present invention obvious. None of these other documents shows the distinguishing features, or provides any according hints. None of these documents shows a set of random system states being generated based on a predetermined probability distribution.

4.4.3 D1 + common knowledge

Also a combination of D1 and the common knowledge of the skilled person fails to render the present invention obvious. As the examiner readily knows, common knowledge can only be relied upon for dismissing inventive step, if said common knowledge is proven. No documents are on file, which are suitable for proving common knowledge. As the examiner readily knows, only text book excerpts, study manuals and the like can be used for proving common knowledge. No such documents are on file. Therefore, all elaborations with regard to common knowledge are moot.

Especially, the examiner's suggestion in section 1.5, that the features of previous dependent claim 10 which now constitute the distinguishing features were one of several straight forward possibilities is such an argumentation with common knowledge. Since this common knowledge is not proven, the examiner's argumentation is moot.

Inventive step is therefore readily given.

5. Conclusion

The examiner is most respectfully requested to reconsider and to announce the intention to grant a European patent based upon the documents now on file. In case the examiner is not yet convinced, a further substantial office action pointing out any remaining objections in detail is requested. Only as a precautionary measure, in case of an intended rejection of the present application, oral proceedings are requested. In case of oral proceedings being scheduled to take place in Berlin or in Rijswijk, a video conference is requested.

Patent Attorney

Clemens Thun

Enclosures
amended claims 1 – 13 (clean copy and amendments marked)
amended description pages 3, 17 (clean copy)
amended description pages 3, 17, 18 (amendments marked)